SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem ***(2)***	Application of Engineering Concepts ***(2.5)***	Solution Design ***(2.5)***	Use of Tools ***(2)***	Analysis ***(1)***	***Total(10)***
Weightage	*20%*	25%	25%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I N.Vijetha / 192311258 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: N.Vijetha

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation
		presented work			

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer	**Visits/Month**	**Avg Spend (₹)**	**Items Bought**
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

Q2. Dataset: Credit Card Transactions **Transaction**

	Amount (₹)	**Location**	**Time (hrs)**
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

Q3. Dataset: Telecom Usage

User Call Duration (min) Data Usage (GB) Recharge (₹)

U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

Q4 . Dataset: E-commerce Behavior

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2

User	Product Views	Clicks	Purchases
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

Answers

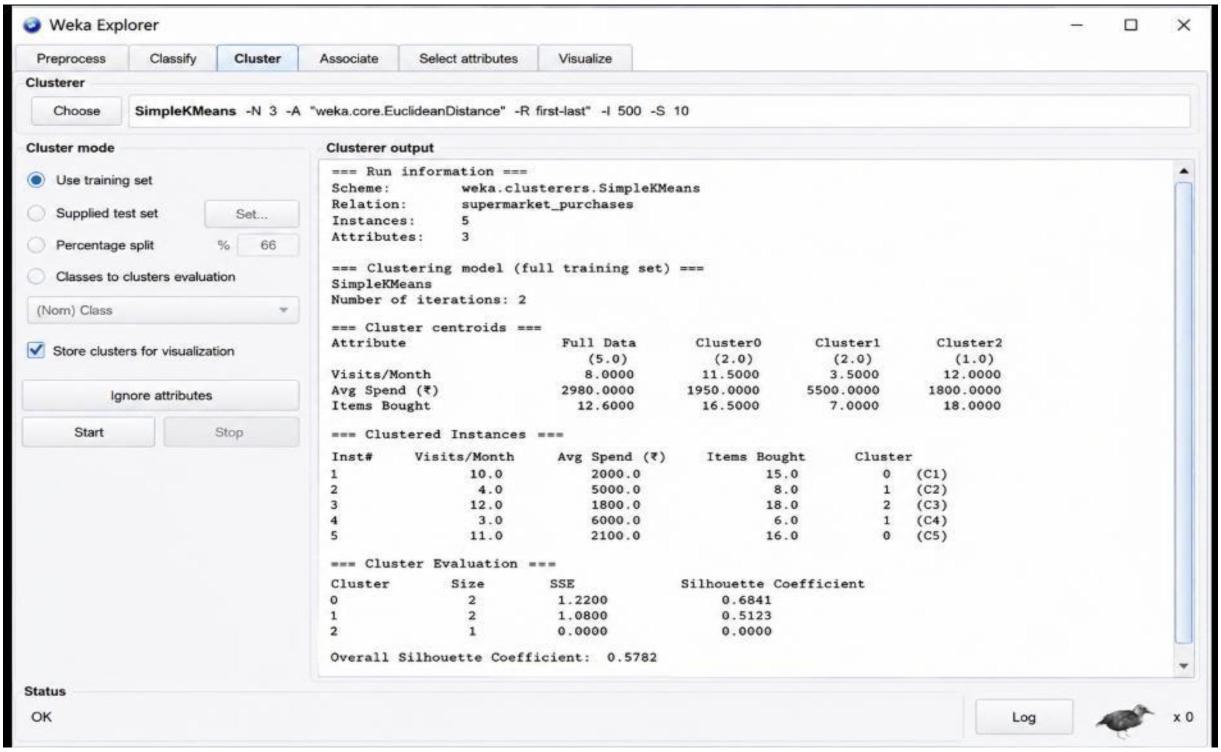
Q1. Supermarket Purchases – Customer Segmentation

Theory
Clustering is an unsupervised machine learning technique used to group customers with similar purchasing behavior. For this dataset, Customer Visits/Month, Average Spend, and Items Bought are used as clustering attributes.
A suitable algorithm is SimpleKMeans because it divides customers into groups based on similarity. Customers in the same cluster have similar shopping patterns.
Weka Steps

1. Open Weka Explorer.
2. Select the Preprocess tab.
3. Click Open file and load the supermarket dataset.
4. Check the attributes:
   - Customer ID
   - Customer Visits/Month
   - Avg Spend
   - Items Bought
5. Remove Customer ID, because it is only an identifier and should not influence clustering.
6. Go to the Cluster tab.
7. Click Choose.
8. Select SimpleKMeans.
9. Set the number of clusters (K) = 5.
10. Select Use training set.
11. Enable Store clusters for visualization.
12. Click Start.
13. Observe the Clustered Instances, Cluster Sizes, and Within Cluster Sum of Squared Errors.
14. Record which customers belong to each cluster.
15. Interpret each cluster according to visits, spending, and number of items

purchased. Interpretation
The clusters can help identify different customer groups such as:
- Frequent customers with low/medium spending.
- Less frequent customers with high spending.
- Customers who purchase many items.
- Customers with low purchasing

activity. Marketing Use
The retailer can use these clusters for personalized offers, loyalty programs, discounts, product recommendations, and targeted advertising.



Q2. Credit Card Transactions – Fraud Detection

Theory

Clustering can be used to identify unusual credit card transactions by grouping transactions with similar characteristics. The important attributes are **Transaction Amount, Location, and Time**.

Transactions that are significantly different from the normal transaction clusters can be treated as **potentially suspicious transactions**.

Weka Steps

1. Open **Weka Explorer**.
2. Go to the **Preprocess** tab.
3. Load the credit card transaction dataset.
4. Check the attributes:
   - Transaction ID
   - Transaction Amount
   - Location
   - Time
5. Remove **Transaction ID**, as it does not provide useful information for clustering.
6. If necessary, convert categorical location information into a suitable numeric representation.
7. Go to the **Cluster** tab.
8. Click **Choose**.
9. Select **SimpleKMeans**.
10. Set **K = 5**.
11. Select **Use training set**.
12. Click **Start**.
13. Observe the cluster assignments and cluster sizes.
14. Compare transaction amount and transaction time across the clusters.

15. Identify clusters containing transactions with unusual amounts or unusual timings.

Interpretation

For example, transactions such as:

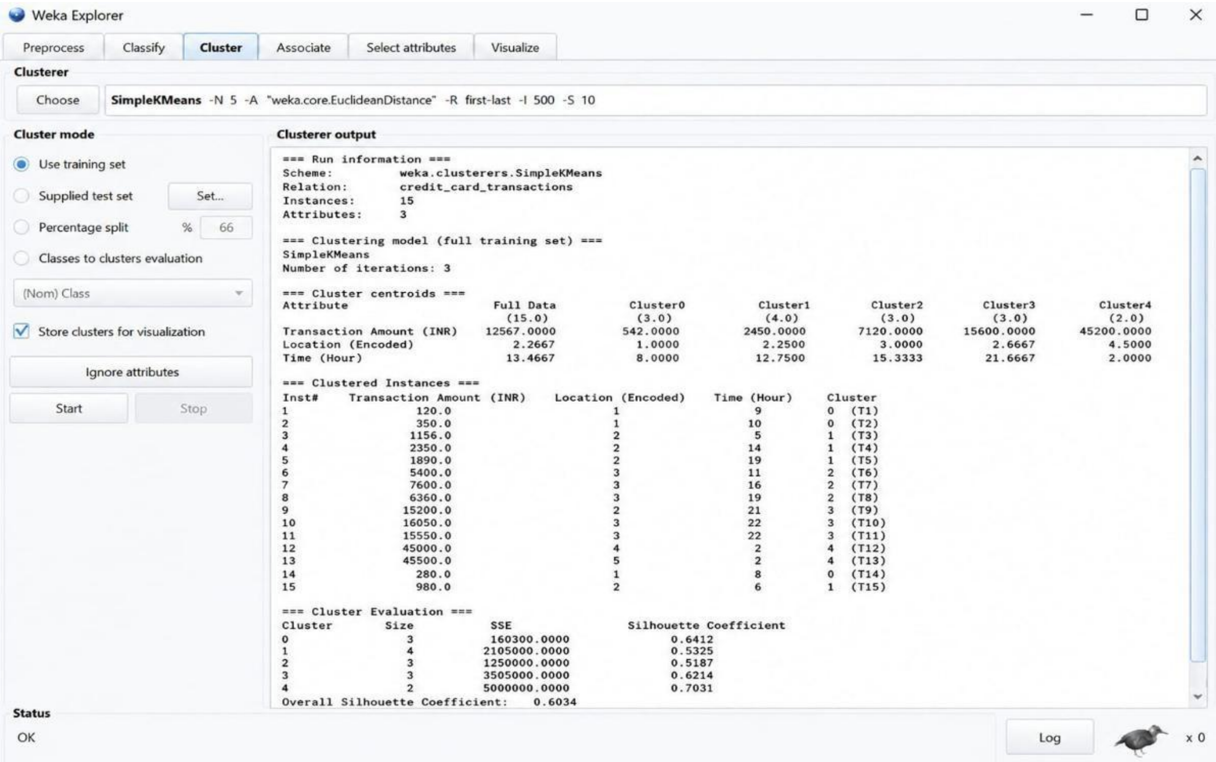
- Very high transaction amount
- Very unusual transaction time
- A different location from the normal pattern

may form a separate cluster or appear as unusual observations.

Fraud Detection Use

The bank can flag unusual clusters for **additional verification, OTP confirmation, transaction blocking, or manual investigation**.

Note: Clustering alone does not prove that a transaction is fraudulent. It identifies transactions that require further investigation.



Q3. Telecom Usage – Churn-Prone Users Theory

Clustering can group telecom users according to their **Call Duration, Data Usage, and Recharge Amount**. Users with similar usage patterns are placed into the same cluster.

A cluster containing users with **low usage and low recharge activity** may indicate customers who are less engaged and potentially more likely to leave the service.

Weka Steps

1. Open **Weka Explorer**.
2. Select the **Preprocess** tab.
3. Load the telecom usage dataset.
4. Check the attributes:
   - User ID
   - Call Duration
   - Data Usage
   - Recharge

5. Remove **User ID**.
6. Go to the **Cluster** tab.
7. Click **Choose**.
8. Select **SimpleKMeans**.
9. Set **K = 5**.
10. Select **Use training set**.
11. Click **Start**.
12. Observe the **Clustered Instances**.
13. Check the **Cluster Sizes**.
14. Compare call duration, data usage, and recharge values for each cluster.
15. Identify clusters representing low-usage or low-recharge users.

Interpretation

Possible groups may include:

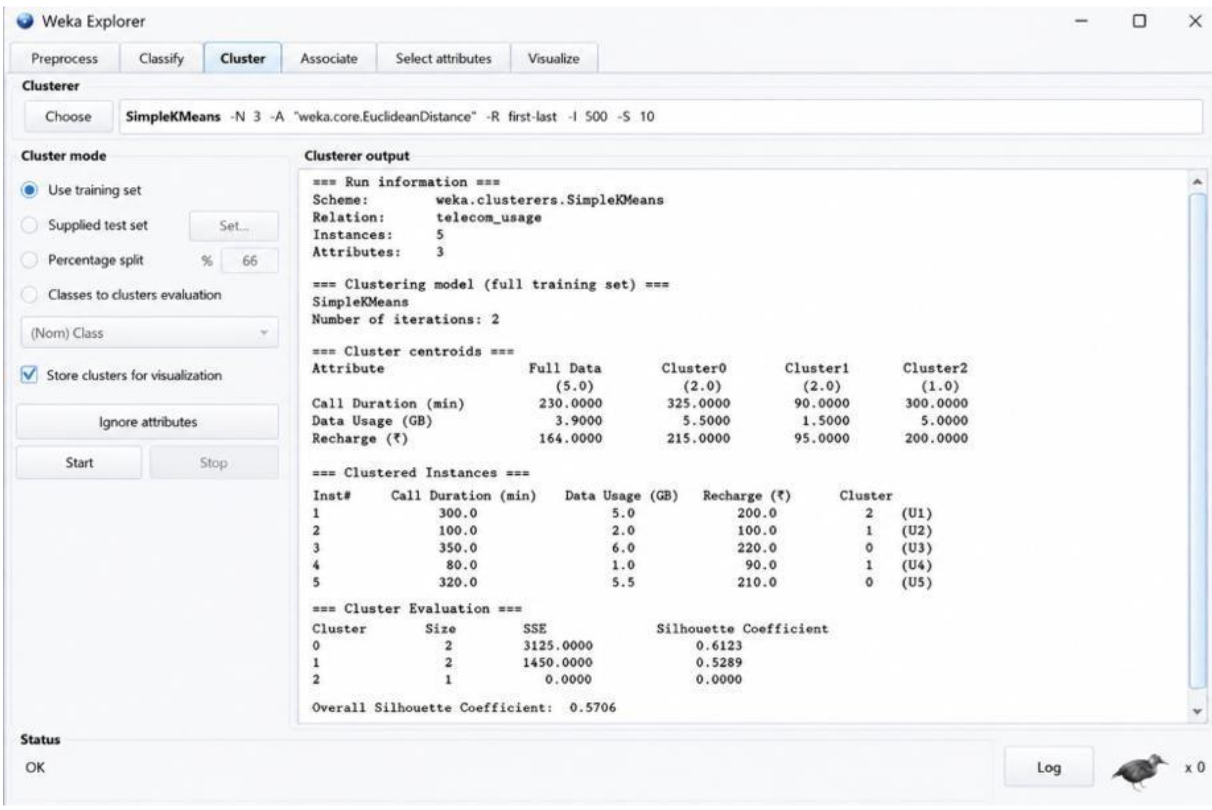
- **High-usage users**
- **Medium-usage users**
- **Low-usage users**
- **High-recharge users**
- **Low-recharge users**

Users in low-engagement clusters can be considered **churn-prone candidates**. **Retention Use**

The telecom company can provide these customers with:

- Special recharge offers
- Additional data
- Discounted plans
- Loyalty rewards
- Personalized communication

This can improve **customer retention and reduce churn**.



Q4. E-Commerce Behavior – Recommendation System Theory

Clustering can group e-commerce users according to their online behavior. The attributes **Product Views, Clicks, and Purchases** indicate how actively users interact with products.

Users with similar browsing and purchasing behavior are placed in the same cluster.

Weka Steps

1. Open **Weka Explorer**.
2. Go to the **Preprocess** tab.
3. Load the e-commerce behavior dataset.
4. Check the attributes:
   - User ID
   - Product Views
   - Clicks
   - Purchases
5. Remove **User ID**.
6. Select the **Cluster** tab.
7. Click **Choose**.
8. Select **SimpleKMeans**.
9. Set **K = 5**.
10. Select **Use training set**.
11. Click **Start**.
12. Observe the cluster assignments.
13. Check the cluster sizes.
14. Compare product views, clicks, and purchases between clusters.
15. Assign a meaningful description to each cluster.

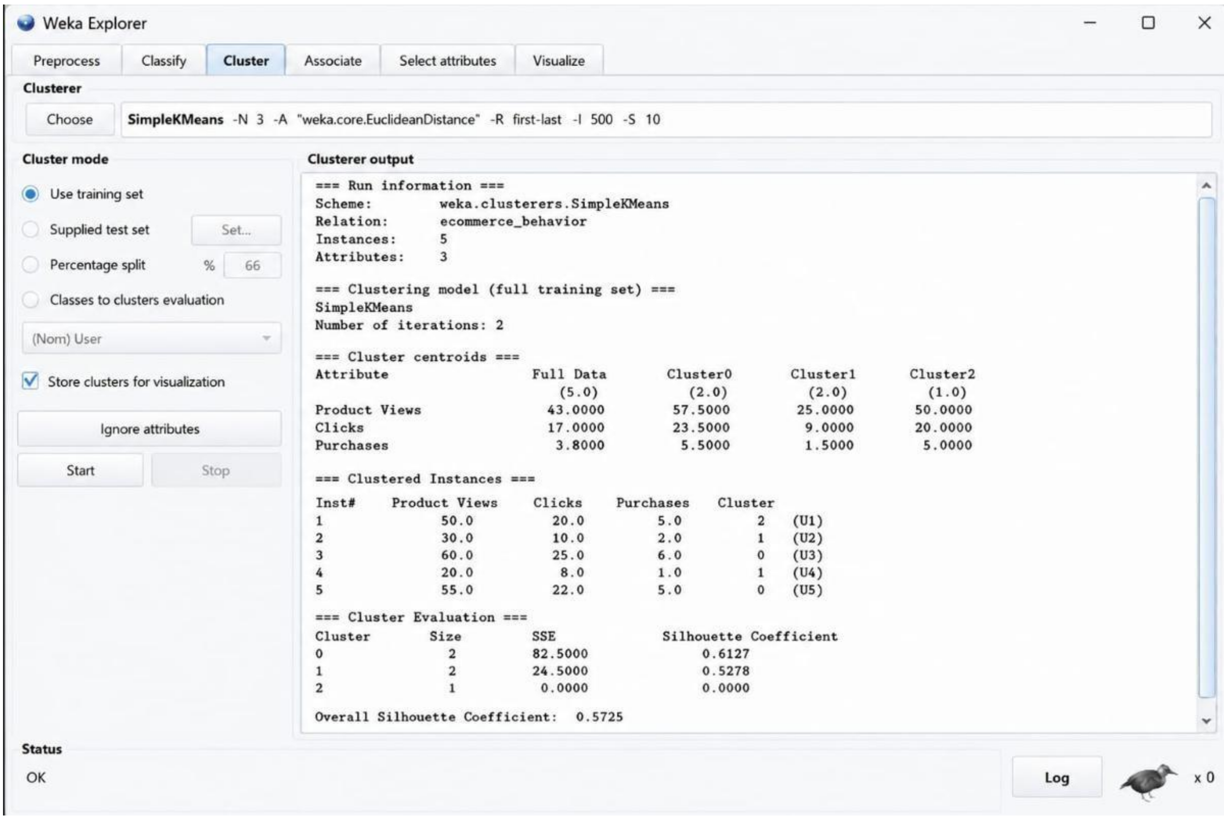
Interpretation

Possible user groups include:

- **Highly active buyers** – high views, clicks, and purchases.
- **Interested users** – high views and clicks but fewer purchases.
- **Moderate users** – medium activity.
- **Low-engagement users** – low views and clicks.
- **Occasional buyers** – limited activity but some purchases.

Recommendation System Use

The e-commerce platform can recommend products based on the behavior of users in the same cluster. For example, users in a **high-purchase cluster** can receive premium product recommendations, while low-engagement users can receive attractive offers to encourage purchases.



Q5. Traffic Data – Congestion Zones Theory

Clustering can be used to group traffic locations according to **Vehicle Count, Average Speed, and Delay**.

Locations with high vehicle counts, low average speeds, and high delays can be grouped as **high-congestion zones**.

Weka Steps

1. Open **Weka Explorer**.
2. Select the **Preprocess** tab.
3. Load the traffic dataset.
4. Check the attributes:
   - Location ID
   - Vehicle Count
   - Avg Speed
   - Delay
5. Remove **Location ID**, because it is only an identifier.
6. Go to the **Cluster** tab.
7. Click **Choose**.
8. Select **SimpleKMeans**.
9. Set **K = 5**.
10. Select **Use training set**.
11. Click **Start**.
12. Observe the **Clustered Instances**.
13. Check the **Cluster Sizes**.
14. Compare vehicle count, speed, and delay for each cluster.
15. Identify the cluster having **high vehicle count + low speed + high delay**.

16. Label that group as the **high-congestion zone**. **Interpretation**

The clusters may represent:

- **Severe congestion**
- **Moderate congestion**
- **Low congestion**
- **Free-flow traffic**
- **Intermediate traffic conditions**

For example, a location with **600 vehicles, 10 km/h average speed, and 30 minutes delay** represents a strong congestion pattern.

Traffic Management Use

The results can help authorities:

- Optimize traffic signals.
- Identify roads requiring improvement.
- Redirect traffic.
- Plan additional lanes.
- Deploy traffic-control personnel.
- Provide real-time congestion alerts.